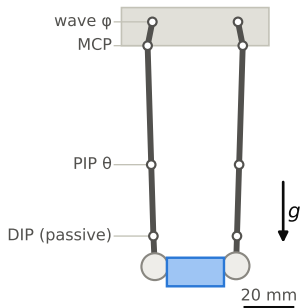
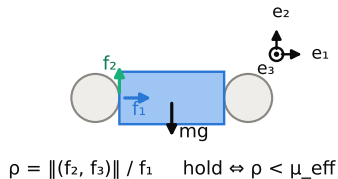


(a) pinch testbed



(b) grasp frame & utilization



(c) thin grasp manifold ($\phi = 0, \theta = 0$)

